

SMART GLOVE FOR SIGN LANGUAGE TRANSLATION USING IoT

PROJECT REPORT

Submitted to

Visvesvaraya Technological University

BELAGAVI - 590 018

by

RACHANA JALISATGI

4SU22EC074

RASHMI SHETTY

4SU22EC082

Under the guidance of

Mrs. Shaileshwari S

Assistant Professor

In partial fulfilment of the requirements for the award of the degree of

Bachelor of Engineering



Department of Electronics & Communication Engineering

SDM INSTITUTE OF TECHNOLOGY

UJIRE – 574240

2025-2026

SDM INSTITUTE OF TECHNOLOGY

(Affiliated to Visvesvaraya Technological University, Belagavi)

UJIRE – 574 240

Department of Electronics & Communication Engineering

CERTIFICATE

Certified that the Project Work titled ‘**Smart Glove for Sign Language Translation using IoT**’ is carried out by **Ms. RACHANA JALISATGI, 4SU22EC074, Ms. RASHMI SHETTY, 4SU22EC082**, are bona-fide students of SDM Institute of Technology, Ujire, in partial fulfilment for the award of the degree of Bachelor of Engineering in Electronics and Communication Engineering of Visvesvaraya Technological University, Belagavi during the year 2025-2026. It is certified that all the corrections/ suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

Mrs. Shaileshwari S
Assistant Professor

Dr. Madhusudhana K
Head of the Department

Dr. Ashok Kumar T
Principal

External Viva

Name of the Examiners:

Signature with Date

1.

2.

Acknowledgement

It is our pleasure to express our heartfelt thanks to **Mrs. Shaileshwari S**, Assistant Professor, Department of Electronics and Communication, for his supervision and guidance which enabled us to understand and develop this project.

We are indebted to **Dr. Madhusudhana K**, Head of the Department and **Dr. Ashok Kumar T**, Principal, for their advice and suggestions at various stages of the work. We also extend our heartfelt gratitude to **Dr. Prathapchandra** and **Dr. Bharathraj Kumar M**, Project Coordinator for his assistance and the management of SDM Institute of Technology, Ujire, for providing us with a good learning environment, library, and laboratory facilities.

Lastly, we take this opportunity to offer our regards to all of those who have supported us directly or indirectly in the successful completion of this project work.

Rachana Jalisatgi
Rashmi Shetty

Abstract

Communication barriers between the hearing- and speech-impaired community and non-sign language users create significant challenges in daily interactions. This project presents a smart glove for sign language translation using IoT, designed to bridge this gap by converting hand gestures into text and speech in real time. The glove is embedded with flex sensors, an accelerometer, a gyroscope, and a microcontroller to detect finger movements and hand orientation. The collected data is processed and transmitted via IoT to a mobile application or display system, where it is translated into readable text and audible speech using a text-to-speech (TTS) module.

The integration of IoT technology allows for real-time wireless communication and cloud-based storage for enhanced accessibility and future improvements. This smart glove provides an efficient, portable, and cost-effective solution for improving communication for the speech and hearing impaired, fostering greater social inclusion.

Table of Contents

	Page No.
Abstract	i
Table of Contents	ii
List of Figures	iv
List of Tables	v
List of Acronyms and Abbreviations	vi
Chapter 1 Introduction	1
Chapter 2 Literature Review	3
2.1 General Introduction	3
2.2 Literature Survey	3
2.3 Summary	13
Chapter 3 Problem Statement and Objectives	14
3.1 Problem statement	14
3.2 Objectives	14
Chapter 4 Methodology	15
4.1 Block diagram	15
4.2 Flow chart	17
4.3 Circuit diagram	18
4.4 Design of Flex sensor	19
4.5 Hardware Components	22
4.6 Software Requirements	28
Chapter 5 Result and Analysis	32
5.1 Result analysis	32
5.2 Summary	36
Chapter 6 Conclusion and Scope for future work	37
6.1 Conclusion	37
6.2 Scope for future work	37
Reference	38

Appendix	40
Plagiarism Report	48
Personal Profile	49

List of Figures

Figure No	Description	Page No.
Figure 1.1	Sign language for deaf and dumb people	1
Figure 1.2	Sign Language chart	2
Figure 4.1	Block diagram	15
Figure 4.2	Flow chart	17
Figure 4.3	Circuit diagram	18
Figure 4.4	Flex sensor circuit	19
Figure 4.5	Flex sensor design	20
Figure 4.6	Arduino Nano	23
Figure 4.7	Arduino Nano pinout	24
Figure 4.8	Flex Sensor	25
Figure 4.9	Accelerometer	26
Figure 4.10	IoT module	27
Figure 4.11	Power Supply	28
Figure 4.12	Arduino IDE	29
Figure 4.13	Mobile App	30
Figure 5.1	Project Setup	32
Figure 5.2	Hand gesture “K” detection and Output display	33
Figure 5.3	Hand gesture “L” detection and Output display	33
Figure 5.4	Hand gesture “N” detection and Output display	34
Figure 5.5	Hand gesture “F” detection and Output display	34
Figure 5.6	Hand gesture “B” detection and Output display	35

List of Tables

Table No	Description	Page No
Table 1	Calibration table for designed Flex sensor	21
Table 2	Features and Specification of Flex sensor	26
Table 3	Features and Specification of Accelerometer	27
Table 4	Result table	35
Table 5	Comparison table	36

List of Acronyms and Abbreviations

CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
ASR	Automatic speech recognition
ISL	Indian Sign Language
ASL	American Sign Language
IMU	Inertial measurement unit
IoT	Internet of things

INTRODUCTION

In everyday life, communication plays a vital role in connecting people, yet many individuals are unfamiliar with sign language the primary means of communication for those who are deaf or mute. Unfortunately, most people are neither trained nor willing to learn it, which often leaves deaf and mute individuals struggling to express themselves or be understood. This communication gap can lead to frustration, isolation, and missed opportunities in daily interactions. To overcome this challenge, we propose a project designed to make communication between nonvocal and vocal individuals easier, more natural, and more inclusive for everyone involved.

Sign language can be understood through three main elements: hand shapes held in place, hand movements, and facial expressions. Among these, hand-based gestures play the most important role in identifying signs. As a result, two people may sign the same message in slightly different ways. This variation is similar to handwriting just as everyone writes the same word differently, sign language also reflects personal style, leading to noticeable differences in how signs are expressed [2].



Figure 1.1: Sign gesture for deaf and dumb people

Figure 1.1 illustrates a sign-gesture system using a smart glove designed to assist individuals with speech impairments in communicating through hand gestures. Using single-handed sign language, the user forms alphabet gestures, which are captured by the glove. The system records these gestures and then translates them into both visual text and audible speech, making communication easier and more accessible.

In this project, flex sensors play a crucial role. These sensors change their electrical resistance based on how much they are bent, allowing the system to accurately detect finger movements. When combined with an accelerometer, the sensors can track not only individual finger motions but also the movement of the entire hand and palm, ensuring precise recognition of gestures [1].

Speech-to-sign language translation involves several key steps. First, the Google Speech API is used to convert the user's spoken words into text, accurately capturing the speech input. This text is then analyzed to identify words, phrases, or complete sentences that can be represented using sign language gestures. The processed text is compared with a database of predefined sign language gestures, where each spoken language element is mapped to its corresponding sign. To ensure accurate and reliable recognition, advanced machine learning models such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) are commonly employed. These models are well-suited for handling large datasets and adapting to variations in individual signing styles [4].

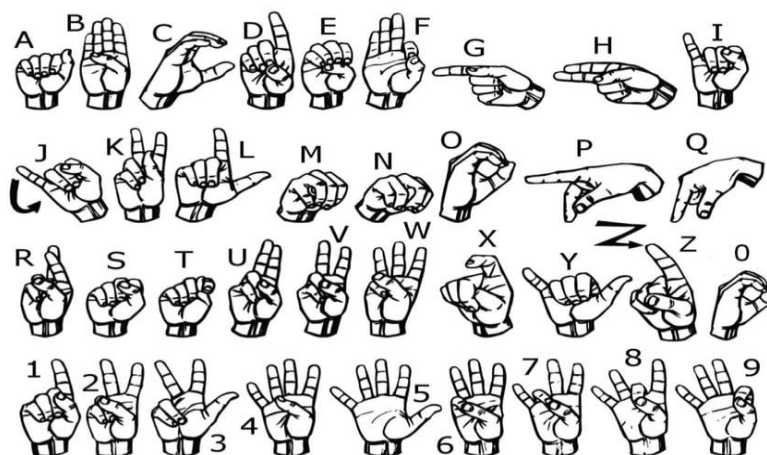


Figure 1.2 Sign Language chart

The sign-to-speech converter is designed to help individuals with speech and hearing impairments communicate more easily. Using smart gloves equipped with sensors and an Arduino Nano, the system captures hand gestures and converts them into spoken words. This technology bridges the communication gap between sign language users and others, promoting independence and inclusivity. With IoT integration, it can also support remote communication. Overall, the goal of this project is to develop smart gloves that translate sign language into speech, improving everyday interactions for the differently abled.

LITERATURE REVIEW

2.1 General introduction:

This literature review explores existing research on wearable technology for sign language translation. It highlights the integration of sensors and IoT for real-time gesture recognition and communication. The review also identifies key advancements and challenges in developing accurate, user-friendly systems.

2.2 Literature Survey

2.2.1 Smart Speaking Glove for Deaf and Dumb

The paper titled “Smart Speaking Glove for Deaf and Dumb” was authored by P. Mohan, M. Mohan Raj, M. Kathirvel, P. A. Kasthurirngan, S. Musharaff, and T. Nirmal Kumar. This paper introduces an innovative assistive device a smart glove designed to help individuals with speech and hearing impairments communicate more effectively. The glove uses flex sensors to detect finger movements and gestures, which are then interpreted by an ATmega328 microcontroller. The identified gestures are translated into both visual texts displayed on an LCD screen and audible speech output via a speaker. The system operates in real time and is programmed to recognize a set of predefined hand gestures corresponding to common phrases or words. Additionally, the researchers suggest that the glove could be extended to control home appliances, increasing its utility in daily life. This device aims to bridge the communication gap for people with hearing and speech disabilities, promoting greater inclusivity and independence [1].

2.2.2 Sign Language to Speech Translation

The paper titled “Sign Language to Speech Translation” was authored by Aishwarya Sharma, Dr. Siba Panda, and Prof. Saurav Verma. The paper outlines a system designed to translate “Indian Sign Language (ISL)” into spoken English in real-time. The primary aim of the study is to bridge the communication gap between individuals with hearing or speech impairments and those who do not understand sign language. The system operates by capturing hand gestures through a camera, preprocessing the images using techniques such as segmentation and background subtraction, and recognizing gestures using machine learning algorithms. Once the gesture is identified, it is converted into

corresponding text, which is then spoken aloud using text-to-speech technology. The solution proposed in this paper emphasizes low-cost implementation and real-time performance, making it suitable for use in public spaces, educational settings, and personal communication tools. The research contributes significantly to the field of assistive technologies by promoting inclusivity and accessibility for the differently-abled community [2].

2.2.3 Design and Implementation of a sign-to speech/text system for deaf and dumb people

The paper titled “Design and Implementation of a Sign-to-Speech/Text System for Deaf and Dumb People” was authored by Dalal Abdulla, Shahrazad Abdulla, Rameesa Manaf, and Anwar H. Jarndal. This research focuses on developing a “smart glove system” aimed at facilitating communication for individuals who are deaf and mute. The system utilizes “gesture recognition” to translate sign language into “Arabic text and speech”, thereby bridging communication gaps between non-verbal individuals and the hearing community. The glove is interfaced wirelessly with a microcontroller and connected to devices that display text or produce speech. The system has been implemented, programmed, cased, and tested, yielding positive results [3].

2.2.4 Real Time Sign Language Translator Using Machine Learning

The paper titled “Real Time Sign Language Translator Using Machine Learning” was authored by Pradnya Repal, Nikita Pore, Akanksha Pasale, Rutuja Thengal, Swati Ambule, and Prof. N.D. Deval. The study presents a real-time, web-based sign language translation system developed using machine learning techniques to assist individuals with hearing and speech impairments. The system captures hand gestures using a webcam and processes the images through machine learning models built using TensorFlow to recognize and classify the gestures. These gestures are then converted into readable text and further transformed into audible speech using text-to-speech synthesis. The application aims to provide a seamless two-way communication tool, capable of translating both sign language into spoken/written language and vice versa. The project stands out for its versatility, supporting multiple signs and spoken languages, and its practical, user-friendly design that enhances real-time interaction between hearing-impaired users and the wider community. This work is a significant contribution to inclusive technology and digital accessibility [4].

2.2.5 Smart Glove for Sign Language Communications

The paper titled "Smart Glove for Sign Language Communications" and authored by Abhinandan Das and colleagues. This research focuses on developing a wearable smart glove designed to translate sign language gestures into text or speech, thereby facilitating communication for individuals with hearing and speech impairments.

The smart glove integrates flex sensors along each finger to detect finger bending and a MEMS accelerometer to capture hand orientation and movement. These sensors work in tandem to interpret the dynamic gestures of sign language. The collected data is processed using a Recurrent Neural Network (RNN) model, which classifies the gestures into corresponding text or speech outputs. The system was trained on a dataset comprising 27 different classes, enhancing its ability to recognize a diverse set of American Sign Language (ASL) gestures. The study demonstrates the potential of combining sensor technology with machine learning algorithms to create an effective real-time sign language translation tool. Such innovations aim to bridge communication gaps between the hearing and non-hearing communities, promoting inclusivity and accessibility [5].

2.2.6 Machine Learning-Based Gesture Recognition Glove: Design and Implementation

The paper titled "Machine Learning-Based Gesture Recognition Glove: Design and Implementation" by Giovanni Pezzuto, Carmine Zaccagnino, and Giuseppe Valerio Lettieri, presents the development of a low-cost smart glove designed to capture and classify dynamic hand gestures using machine learning techniques. The glove is equipped with five flex sensors, five force sensors, and an inertial measurement unit (IMU), all connected to an ESP32-S2 microcontroller. Data was collected at 100 Hz from multiple participants performing five different gestures. The authors designed and trained a convolutional neural network (CNN) with three convolutional layers to classify these gestures, achieving an accuracy of 90% and an F1-score of 0.9132. The study emphasizes the glove's potential applications in virtual and augmented reality, game control, and assistive technologies for people with disabilities. While the research was limited by the small number of gestures and participants, the results are promising, and the authors suggest future work to improve glove comfort, increase gesture variety, and test the system in realworld scenarios [6].

2.2.7 An IoT-Based Smart Glove Gesture Vocalizer for Deaf and Mute People

The paper titled "An IoT-Based Smart Glove Gesture Vocalizer for Deaf and Mute People", authored by Jyothi R. and Lakshmishri B. L. This study addresses the communication challenges faced by individuals who are deaf and mute by developing a smart glove that translates sign language gestures into audible speech. The proposed system employs flex sensors embedded in a glove to detect hand movements corresponding to specific sign language gestures. These sensor readings are processed by a microcontroller, which interprets the gestures and converts them into vocalized speech using a text-to-speech module. The integration of Internet of Things (IoT) technology allows for real-time communication, enabling users to interact more effectively with others in various environments, including educational, social, and professional settings. This innovation aims to bridge the communication gap between the differently-abled community and the general public, promoting inclusivity and enhancing the quality of life for individuals with hearing and speech impairments [7].

2.2.8 Multi-Purpose Smart Glove for Differently Abled Community People

The paper titled "Multi-Purpose Smart Glove for Differently Abled Community People" authored by Mr. Punit Vora, Mr. Bhushan Parab, Mr. Rohit Bandgar, Mr. Rahul Naik, Dr. Jyoti Dange. This presents an innovative wearable device designed to assist individuals with disabilities, particularly those who are deaf or hard of hearing. The primary objective of this smart glove is to bridge the communication gap between differently-abled individuals and the general public by translating sign language gestures into both text and speech formats. This functionality is achieved through the integration of flex sensors that detect finger movements, which are then processed by a microcontroller to interpret specific gestures. The interpreted gestures are subsequently converted into audible speech and displayed as text, facilitating effective communication without the need for an interpreter.

The device is designed to be portable and user-friendly, with an accompanying mobile application that assists users in switching between various operational modes. This multifunctional approach not only addresses communication barriers but also supports daily living activities, making it a comprehensive assistive tool for individuals with special needs [8].

2.2.9 Development of a Smart Glove for Sign Language Translation Using Arduino

The paper titled "Development of a Smart Glove for Sign Language Translation Using Arduino," authored by Ahmed J Abougarair, Walaa A Arebi. This presents a project focused on creating an affordable and efficient wearable device to assist communication for the hearing and speech impaired. The smart glove is equipped with flex sensors on the fingers to detect bending movements, an accelerometer on the palm to sense hand orientation, and touch sensors between the fingers to recognize specific contacts. These sensors capture various hand gestures representing sign language, and the data is processed by an Arduino UNO microcontroller. The processed information is then transmitted via Bluetooth to an Android smartphone, where it is converted into readable text and spoken words using a text-to-speech module [9].

2.2.10 Smart Glove for Sign Language Recognition Using Flex and Accelerometer Sensors

The paper titled "Smart Glove for Sign Language Recognition Using Flex and Accelerometer Sensors," authored by Yasir Niaz Khan, Syed Atif Mehdi. This presents the development of a wearable device aimed at assisting communication for individuals with hearing and speech impairments. The smart glove integrates flex sensors attached to the fingers to measure bending and an accelerometer to detect hand orientation and movement. Together, these sensors capture a wide range of sign language gestures, which are processed by a microcontroller to translate the gestures into corresponding letters or words. The output is then displayed as text or converted to speech via a text-to-speech module, enabling real-time communication. Testing showed the system effectively recognizes various gestures with good accuracy, highlighting the smart glove as a practical and non-invasive solution to bridge communication gaps and improve accessibility for the differently-abled community [10].

2.2.11 EchoGlove: Assistive Smart Glove for Gesture-Based Communication and Health Monitoring

The paper titled "EchoGlove: Assistive Smart Glove for Gesture-Based Communication and Health Monitoring" authored by Vijay A, Lochan Pingalay S, Mohanakrishnan. N, Rohan Raghavendra. G, Shalini Agnes Mary A. This presents a comprehensive wearable solution aimed at enhancing communication and safety for individuals with speech and hearing impairments. Developed by researchers from the International Journal for

Research Trends and Innovation (IJRTI), EchoGlove integrates multiple functionalities into a single, cost-effective device. The glove utilizes an ESP32 microcontroller to process inputs from various sensors: flex sensors detect specific hand gestures, a pulse sensor monitors heart rate, and an ultrasonic sensor identifies nearby obstacles. Recognized gestures are converted into audible speech via an onboard speaker, facilitating real-time communication. Additionally, the system connects to a mobile application through Bluetooth, displaying the interpreted gestures as text and tracking health metrics, thereby promoting user independence and safety.

The integration of health monitoring features, such as heart rate tracking with alerts for irregularities, adds a critical layer of safety. The obstacle detection capability further assists users in navigating their environment confidently. By combining gesture recognition, speech output, health monitoring, and environmental awareness, EchoGlove stands out as a multifaceted assistive technology. Its affordability and comprehensive feature set make it a promising tool for improving the quality of life for those with communication challenges [11].

2.2.12 Sign Language Translation Using Smart Glove and Mobile Application

The paper titled "Sign Language Translation Using Smart Glove and Mobile Application," authored by Ahmed J Abougarair, Walaa A Arebi. This presents a wearable system designed to facilitate real-time translation of sign language into text and speech. The smart glove incorporates five flex sensors attached to the fingers to detect bending motions and an accelerometer to capture hand orientation and movement. These sensors work in tandem to recognize specific gestures corresponding to the American Sign Language (ASL) alphabet. The captured data is processed by a microcontroller and transmitted via Bluetooth to a mobile application, which then converts the gestures into readable text and synthesized speech. Preliminary experiments demonstrated that the system could translate simple ASL signs with an average response time of approximately 0.6 seconds, highlighting its potential for real-time communication. This integration of hardware and software offers a cost-effective and portable solution to bridge the communication gap between individuals who use sign language and those unfamiliar with it [12].

2.2.13 Wearable Glove for Sign Language Recognition Using Flex and Gyroscope Sensors

The paper titled "Wearable Glove for Sign Language Recognition Using Flex and Gyroscope Sensors," authored by Urav Dalal Aasmi Thadhani, Mahek Upadhye, Shreya Shah, Dr. Meera Narvekar, Dr. Nilesh Patil. This presents the design and implementation of a smart glove aimed at translating sign language gestures into text and speech. The glove integrates five flex sensors, each attached to a finger, to detect bending motions, and an MPU-6050 module that combines an accelerometer and a gyroscope to capture hand orientation and movement. These sensors work in tandem to recognize specific gestures corresponding to the American Sign Language (ASL) alphabet.

The captured data is processed by a microcontroller and transmitted via Bluetooth to a mobile application, which then converts the gestures into readable text and synthesized speech. Preliminary experiments demonstrated that the system could translate simple ASL signs with an average response time of approximately 0.6 seconds, highlighting its potential for real-time communication. This integration of hardware and software offers a cost-effective and portable solution to bridge the communication gap between individuals who use sign language and those unfamiliar with it [13].

2.2.14 Real-Time Sign Language Translation Using Wearable Sensors and Deep Learning Technique

The paper titled "Real-Time Sign Language Translation Using Wearable Sensors and Deep Learning Technique," authored by “This presents a novel system that combines wearable sensor technology with advanced deep learning models to achieve real-time sign language translation. The system features a glove embedded with multiple sensors, including flex sensors to capture finger bending and an accelerometer to detect hand motion and orientation. These sensors generate continuous gesture data, which is fed into a deep learning framework composed of convolutional neural networks (CNNs) and recurrent neural networks (RNNs) to classify and interpret sign language gestures with high accuracy.

The research focuses on translating dynamic hand gestures into textual output in real time, aiming to bridge the communication gap for individuals with hearing or speech impairments. Experimental results demonstrate that the proposed system achieves high gesture recognition accuracy, fast processing speed, and practical usability. The integration of deep learning significantly improves the system's ability to learn complex

gesture patterns and adapt to variations in hand movement. Overall, the study contributes a powerful, responsive, and userfriendly solution for inclusive communication using cutting-edge AI and wearable technology [14].

2.2.15 Deep Learning-Enabled Smart Glove for Real-Time Sign Language

Translation

The paper titled "Deep Learning-Enabled Smart Glove for Real-Time Sign Language Translation" by Urav Dalal, Aasmi Thadhani, Mahek Upadhye, Shreya Shah, Meera Narvekar, and Nilesh Patil. This presents the design and development of a wearable smart glove aimed at translating American Sign Language (ASL) gestures into text using deep learning techniques. The glove integrates five flex sensors to capture finger bends and an MPU-6050 sensor to detect hand orientation and motion. These inputs are processed by a Bidirectional Long Short-Term Memory (Bi-LSTM) neural network, which effectively recognizes gesture sequences in real-time. The model is trained on a custom dataset that includes various ASL signs, enabling accurate recognition of both static and dynamic hand movements. The system is designed to aid individuals with speech or hearing impairments by providing a real-time, text-based communication method. Future enhancements include incorporating wireless connectivity, expanding the system to support Indian Sign Language (ISL), and adding a text-to-speech module for voice output. This work demonstrates the promising potential of combining wearable technology with deep learning to bridge communication gaps in inclusive settings [15].

2.2.16 Electronic Speaking Glove for Speechless Patients

The paper titled "Electronic Speaking Glove for Speechless Patients" by Ali, S. Munawwar, and B. Nadeem, focuses on the design and development of a wearable glove system intended to assist speechless patients in communicating. The glove integrates flex sensors to detect finger movements corresponding to specific sign language gestures. These sensor signals are processed through an electronic circuit interfaced with a microcontroller, which translates the gestures into predefined text or speech outputs via a speaker or display unit. The project emphasizes affordability and accessibility, aiming to provide a low-cost assistive device that can enhance the quality of life and communication abilities of individuals with speech impairments. The study highlights the challenges in accurate gesture detection and real-time processing but demonstrates promising results,

showing that electronic gloves can be effective tools for sign language translation and communication aid [16].

2.2.17 Captioning for Deaf and Hard of Hearing People by Editing Automatic Speech Recognition in Real Time

The paper titled "Captioning for Deaf and Hard of Hearing People by Editing Automatic Speech Recognition in Real Time" by M. Wald, addresses the challenge of providing accurate and timely captions for deaf and hard-of-hearing individuals. The study explores a system that improves real-time automatic speech recognition (ASR) by incorporating human editing during live transcription, aiming to reduce errors commonly found in automatic captions. This hybrid approach enhances the readability and reliability of captions, enabling better accessibility for users relying on real-time text to follow spoken communication. The paper discusses the technical implementation of the system, its impact on user experience, and the potential to improve communication in various settings such as lectures, meetings, and broadcasts. Overall, this research contributes to advancing assistive technologies that support more inclusive communication for people with hearing impairments [17].

2.2.18 Electronic Hand Glove for Speech Impaired and Paralyzed Patients

The paper "Electronic Hand Glove for Speech Impaired and Paralyzed Patients" by N. P. Bhatti, A. Baqai, B. S. Chowdhry, and M. A. Unar, discusses the design and implementation of a wearable electronic glove aimed at assisting speech-impaired and paralyzed patients in communication. The glove is equipped with flex sensors that detect finger movements corresponding to sign language gestures. These signals are processed by a microcontroller to convert gestures into audible speech or text output, providing an effective communication bridge for individuals unable to speak or move freely. The paper emphasizes the device's potential to improve the quality of life and social interaction for patients with disabilities by enabling them to express themselves more easily. Additionally, the authors highlight the system's cost-effectiveness and portability, making it a practical solution for real-world applications, especially in resource-limited settings [18].

2.2.19 Design and Development of a Smart Glove for Assistive Communication

The paper titled "Design and Development of a Smart Glove for Assistive Communication" by Christopher Harris and Samantha Carter, presented at the IEEE International Conference on Rehabilitation Robotics (ICORR), focuses on creating a wearable smart glove aimed at improving communication for individuals with speech and motor impairments. The glove is equipped with an array of sensors, including flex and inertial measurement units (IMUs), to accurately capture hand and finger movements corresponding to sign language gestures. Using advanced signal processing and machine learning algorithms, the system translates these gestures into spoken or written language in real time. The study highlights the integration of wearable technology with rehabilitation robotics principles to provide an intuitive and non-invasive communication aid.

Results demonstrate the device's potential for high accuracy and user comfort, supporting its use in daily life for patients requiring assistive communication solutions. This research contributes to the broader effort of developing accessible, effective technologies for enhancing independence among people with disabilities [19].

2.2.20 Integration of Machine Learning Algorithms in Gesture Recognition for Assistive Devices

The paper titled "Integration of Machine Learning Algorithms in Gesture Recognition for Assistive Devices" by Daniel Robinson and Rachel White, explores the application of advanced machine learning techniques to enhance gesture recognition systems used in assistive devices. The study focuses on improving the accuracy and responsiveness of wearable technologies, such as smart gloves, that interpret hand and finger movements for communication and control purposes. By integrating algorithms like convolutional neural networks (CNNs) and support vector machines (SVMs), the authors demonstrate significant improvements in recognizing complex gestures across diverse users and environments. The research also addresses challenges related to real-time processing and variability in human motion, proposing robust models that adapt to individual differences. Overall, the paper contributes valuable insights into leveraging machine learning to make assistive devices more intuitive and effective for users with physical impairments [20].

2.3 Summary:

The literature shows growing interest in smart gloves for sign language translation. Most designs use flex sensors and accelerometers to detect gestures. However, many lack real-time communication and IoT features. Recent tech advancements support more efficient, portable solutions. This project builds on past work to create an IoT-based, real-time translation glove.

PROBLEM STATEMENT AND OBJECTIVE

3.1 Problem statement:

Lack of widespread sign language understanding creates communication barriers between deaf or hard-of-hearing individuals and the general public, requiring mediation to ensure their rights and messages are conveyed accurately.

3.2 Objective:

- To develop a wearable smart glove capable of detecting hand and finger movements corresponding to sign language gestures.
- To integrate sensor (flex) for accurate real-time gesture recognition.
- To process and translate gestures into readable text or audible speech.

METHODOLOGY

Figure 4.1 shows a wearable, sensor-based system built around an Arduino Nano as the main controller. It uses four flex sensors and an accelerometer to capture hand movements and gestures. The Arduino processes this data and sends it to a mobile application or IoT platform for further use. A power supply module provides energy to the system, making it suitable for applications such as smart gloves, gesture recognition, health monitoring, and assistive communication devices.

4.1 Block diagram

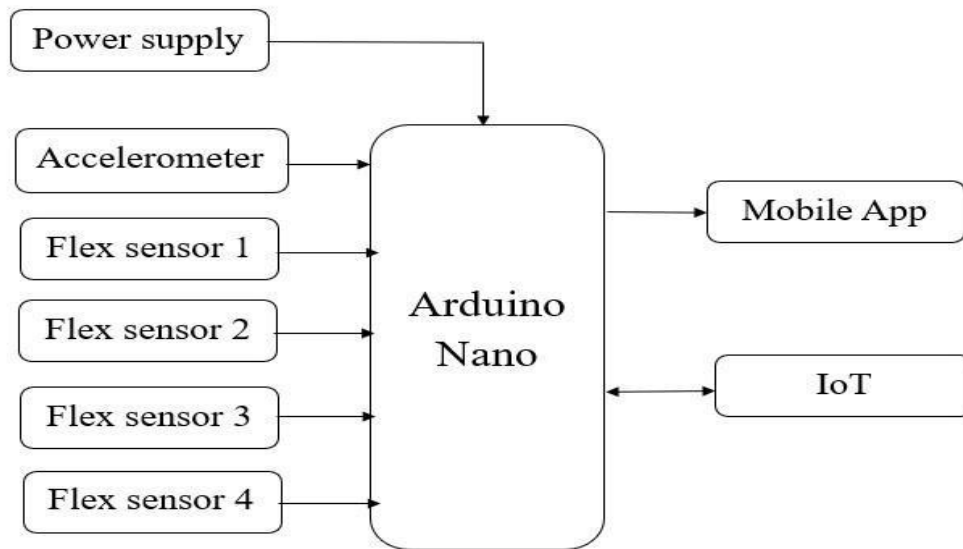


Figure 4.1: Block diagram

Power Supply: Provides the required voltage to all system components.

Accelerometer: Detects hand movement and orientation.

Flex Sensors: Measure the bending of fingers or joints.

Arduino Nano: Acts as the main controller, processing sensor data and managing system operations.

Speaker: Produces audio output based on recognized gestures.

IoT Module: Enables data transmission over the internet for remote access and monitoring.

The development of the Smart Glove for Sign Language Translation is carried out in a series of well-defined stages to ensure smooth operation and reliable performance.

1. System Initialization

- Once power is supplied, the Arduino Nano activates and initializes all connected components, including sensors, communication interfaces, and input/output pins.
- This prepares the system for data collection and processing.

2. Sensor Data Collection

- Flex sensors placed on the fingers continuously measure finger bending and send analog signals to the Arduino.
- At the same time, the accelerometer captures the hand's movement and orientation in three-dimensional space, helping identify different gestures.

3. Data Processing

- The Arduino Nano reads and analyzes data from both the flex sensors and the accelerometer.
- Using predefined logic, the system interprets the sensor values to recognize specific hand gestures or postures.

4. Output Generation

- Once a gesture is recognized, the Arduino triggers appropriate outputs.
- The speaker produces corresponding voice feedback, allowing gestures to be converted into audible speech. Simultaneously, gesture data is prepared for online transmission.

5. IoT Integration

- The IoT module sends real-time gesture data to a cloud server or mobile application. This enables remote monitoring, data storage, and interaction with other smart devices.
- Feedback from the cloud can also be used to control system behaviour.

6. Continuous Monitoring

- The system runs continuously, constantly tracking hand movements, processing sensor data, and generating responses.
- It can also store data or trigger alerts when specific gestures or patterns are detected.

4.2 Flow Chart

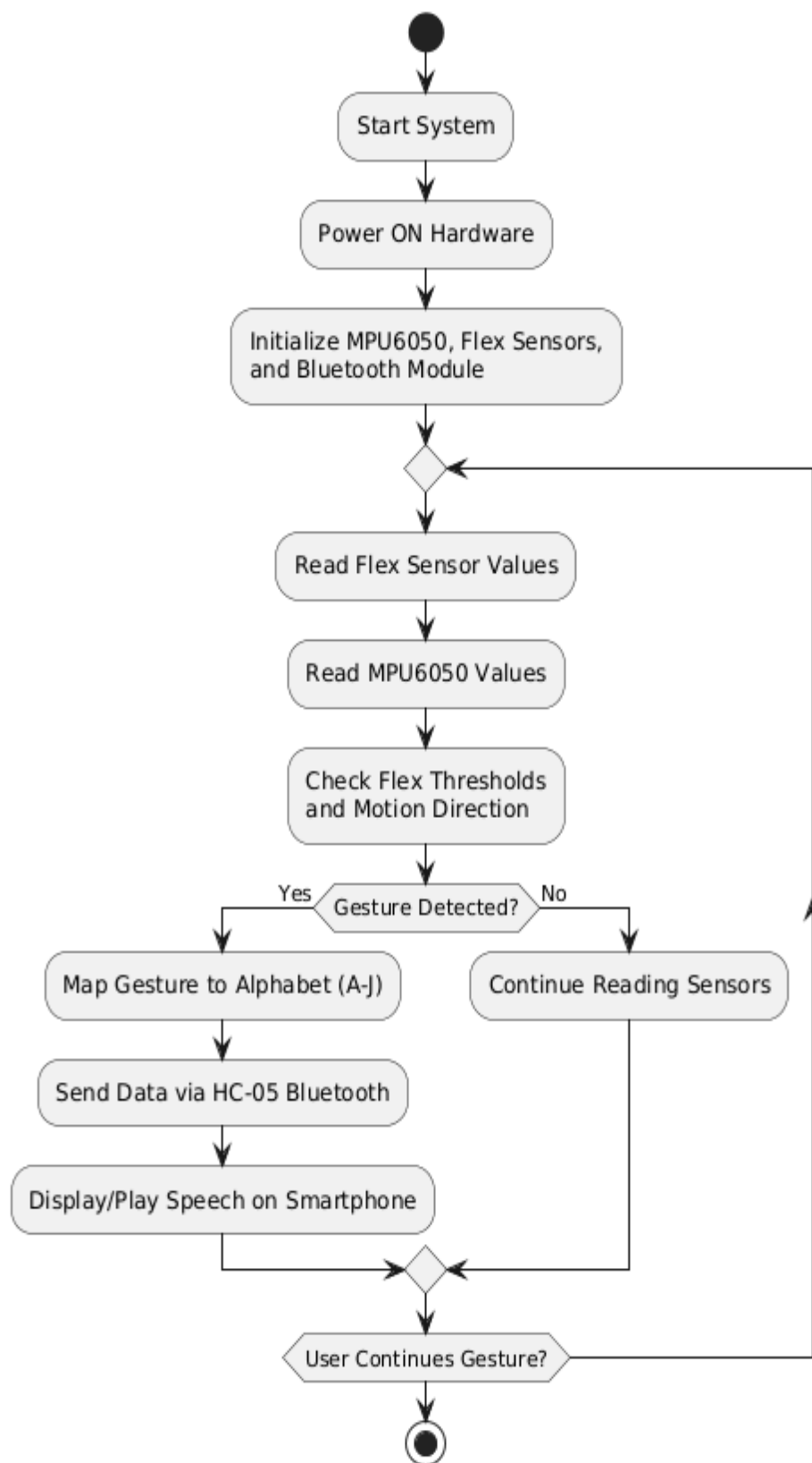


Figure 4.2: Flow chart

The figure 4.2 represents the flow chart of smart glove. The system begins with powering the hardware components. The Arduino initializes the MPU6050 sensor, flex sensors and the Bluetooth module. It continuously reads analog data from flex sensors and directional value from MPU6050. The program checks if the combined values meet preset conditions and identify a gesture. If a gesture pattern matches, the Arduino maps it to an alphabet (A-J). The recognized alphabet is sent wirelessly to a smartphone via Bluetooth. The smartphone application converts the received character to speech output. The process repeats as long as gestures continue to be performed.

4.3 Circuit diagram

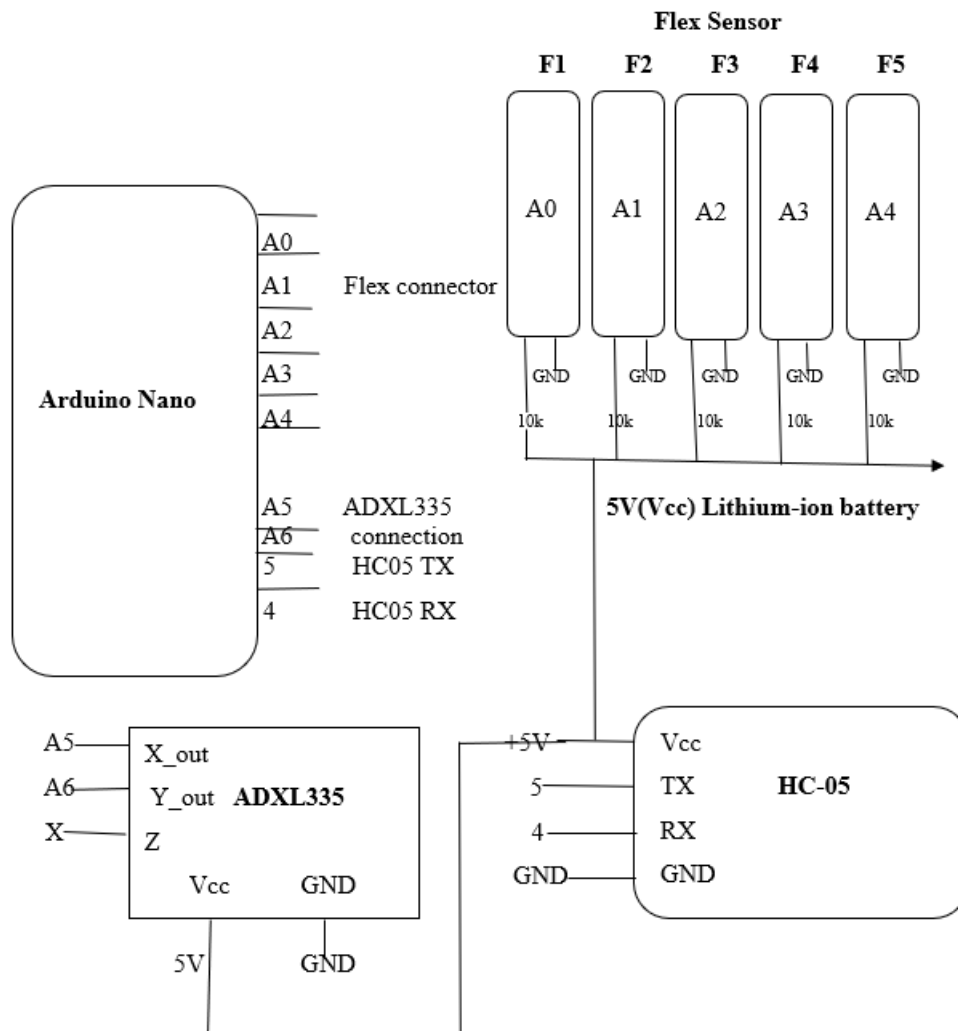


Figure 4.3: Circuit diagram

The figure 4.3 represents a circuit diagram of smart glove for sign language translation using IoT, where the Arduino Nano functions as the main controller. Five flex sensors are connected to the analog input pins A0 to A4 of the Arduino Nano, with each flex sensor forming a voltage divider circuit using a resistor and a +5V supply. When a finger bends, the resistance of the corresponding flex sensor changes, producing a varying analog voltage that is read by the Arduino to detect finger movements. An ADXL335 accelerometer is used to sense hand orientation and motion, with its X and Y output pins connected to analog pins A5 and A6 of the Arduino Nano, while the Z-axis is not utilized. The accelerometer helps in identifying dynamic gestures and improves gesture recognition accuracy. The processed gesture data is transmitted wirelessly using an HC-05 Bluetooth module, where the TX and RX pins are connected to Arduino digital pins 5 and 4 respectively, and powered by a 5V supply. All components share a common ground to ensure stable operation. Overall, this circuit converts hand gestures into electrical signals, processes them using the Arduino Nano, and sends the corresponding message wirelessly, enabling effective communication for deaf and mute individuals.

4.4 Design of Flex Sensor

The flex sensor acts as a variable resistor and is connected with a fixed resistor to form a voltage divider. When the sensor bends, its resistance changes, causing a change in the output voltage. This voltage is passed through an op-amp buffer to provide a stable signal, which is then sent to the microcontroller for gesture detection.

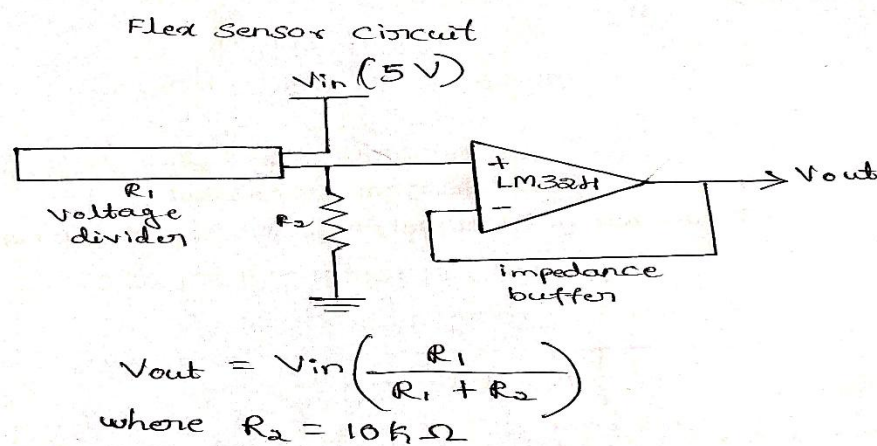


Figure 4.4: Flex sensor circuit

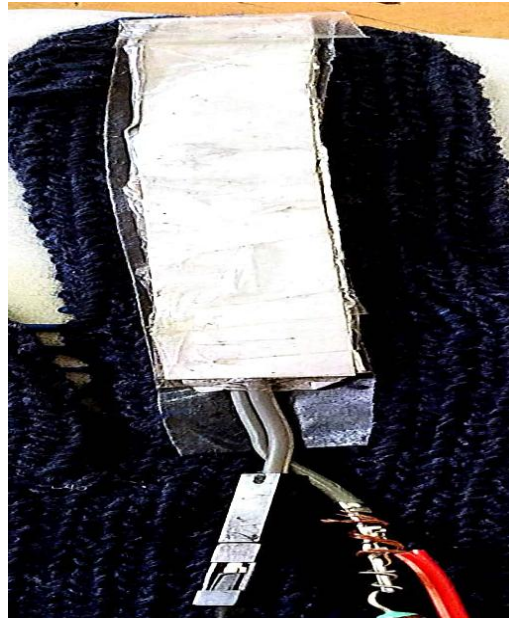


Figure 4.5: Flex sensor design

The flex sensor is made using pencil-sketched paper and aluminium foil. The pencil sketch forms a graphite resistive path whose resistance increases when the paper bends, while the aluminium foil acts as low-resistance terminals for proper electrical connection. This fabricated flex sensor is used as one resistor in a voltage divider circuit with a fixed resistor. When a constant input voltage is applied, bending the sensor changes its resistance and hence the output voltage. This output is passed through an op-amp impedance buffer to provide a stable signal, which is then sent to the microcontroller for gesture detection.

Calculation for designed Flex sensor

Given,

Supply voltage, $V_{in} = 5V$

Fixed resistor, $R = 10k\Omega$

Arduino ADC $\Rightarrow$ 10-bit range 0-1023

Voltage Divider Formula

$$V_{out} = V_{in} \times \frac{R}{R + R_2}$$

ADC Conversion formula

$$ADC = \frac{V_{out}}{V_{in}} \times 1023$$

Bend Angle	Resistance	Voltage	ADC Output in bit/sec
0 ⁰	7 k Ω	2.94	~602
30 ⁰	12 k Ω	2.27	~465
45 ⁰	18 k Ω	1.78	~365
60 ⁰	25 k Ω	1.43	~293
90 ⁰	35 k Ω	1.11	~227

Table 1: Calibration table for designed Flex sensor

Algorithms:

- Threshold-based algorithm - Sensor values from the MPU6050 and flex sensors are compared with predefined threshold limits to detect gestures.
- Rule-based decision algorithm (if-else logic)- Hand gestures are recognized by matching specific sensor conditions using predefined rules.
- Tilt/orientation detection algorithm- Accelerometer data (X and Y axes) is used to determine hand direction based on tilt.
- Polling algorithm- Sensors are continuously read inside the loop without using interrupts.
- Deterministic control algorithm- The system produces fixed outputs for the same sensor inputs, with no learning or adaptation.

4.5 Hardware Components

Main Components

- **Arduino Nano**
Serves as the core controller of the system, responsible for reading sensor data and processing gestures.
- **MPU6050 (Accelerometer and Gyroscope)**
Detects hand movements and directional gestures such as up, down, left, and right, enabling accurate motion tracking.
- **Flex Sensors (4 Units)**
Measure the bending of fingers by producing varying analog voltage signals based on finger movement.
- **IoT Module**
Enables wireless communication between the Arduino Nano and a smartphone for data transmission.
- **Power Supply Module (5V Regulated)**
Ensures a stable and reliable power supply to the Arduino Nano and connected sensors.
- **Lithium-Ion / Rechargeable Battery**
Provides portable power, making the device suitable for wearable applications.

4.5.1 Arduino Nano

Figure 4.6 shows the Arduino Nano, a small yet powerful microcontroller board based on the ATmega328P. Due to its compact size and versatile features, it is widely used in electronics and embedded system projects. The Arduino Nano operates at 5V and provides 14 digital input/output pins, six of which support PWM output, along with eight analog input pins. It also supports standard communication protocols such as UART, SPI, and I2C. The board is equipped with 32 KB of flash memory for program storage, 2 KB of SRAM, and 1 KB of EEPROM, ensuring sufficient memory for most applications. Its 16 MHz clock speed allows for fast and reliable performance. Designed to be breadboard-friendly, the Arduino Nano can be easily programmed and powered through a Mini USB connection.

The Arduino Nano is an ideal choice for applications such as embedded systems, sensor-based projects, robotics, and DIY electronics. Additionally, newer variants like the Nano

Every, Nano 33 IoT, and Nano 33 BLE extend the capabilities of the Nano family by offering modern microcontrollers, built-in Wi-Fi and Bluetooth, and onboard sensors. This makes the Arduino Nano platform suitable for a wide range of projects, from simple LED control to advanced IoT-based applications.

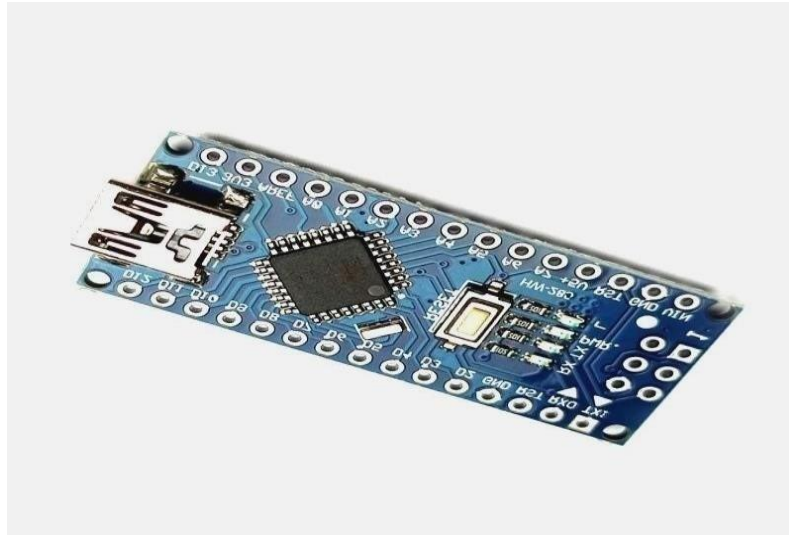


Figure 4.6: Arduino Nano

Arduino Nano Features

The features of an Arduino nano mainly include the following.

- ATmega328P Microcontroller is from 8-bit AVR family
- Operating voltage is 5V
- Input voltage (V_{in}) is 7V to 12V
- Input/Output Pins are 22
- Analog input pins are 6 from A0 to A5
- Digital pins are 14
- Power consumption is 19 Ma
- I/O pins DC Current is 40 mA
- Flash memory is 32 KB
- SRAM is 2 KB
- EEPROM is 1 KB
- CLK speed is 16 MHz
- Weight-7g
- Size of the printed circuit board is 18 X 45mm

- Supports three communications like SPI, IIC, & USART

Arduino Nano Pinout

Arduino nano pin configuration is shown below and each pin functionality is discussed below.

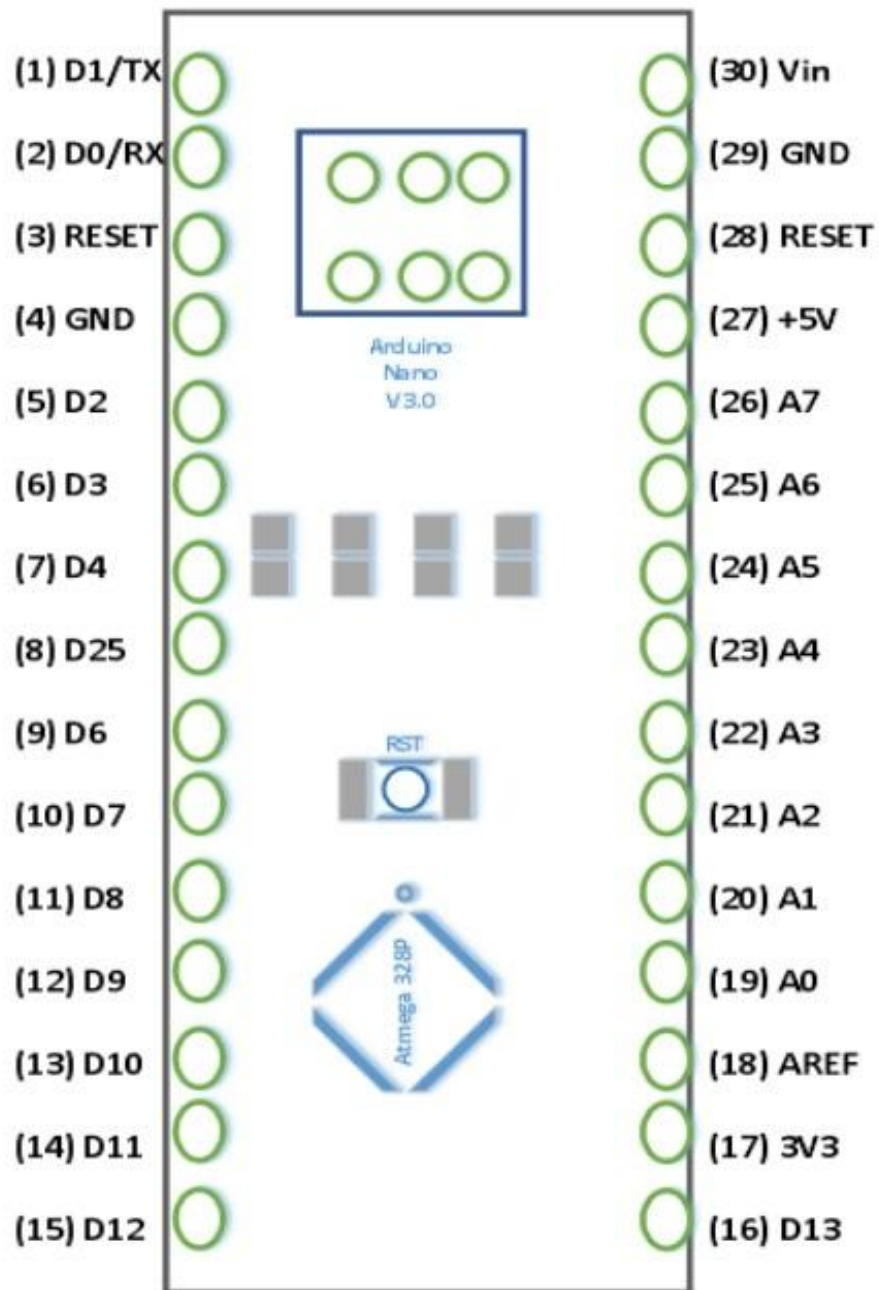


Figure 4.7: Arduino-nano-pinout

4.5.2 Flex Sensor

Figure 4.8 shows flex sensors, which are piezoresistive devices that change their resistance based on the direction and degree of bending. To accurately capture finger movements, a flex sensor is placed on each joint, allowing the system to detect even subtle bends. A custom filter is used to smooth sudden fluctuations, ensuring that the sensor readings remain within acceptable ranges. Since the Arduino's ADC can handle up to 5 volts, a voltage divider circuit with a 5V input is used to maintain consistency between the sensors and the ADC, enabling accurate and reliable gesture detection.

Pin Configuration

The flex sensor has a simple two-terminal configuration, labeled as P1 and P2. Unlike polarized components such as diodes or capacitors, it does not have a positive or negative terminal, so it can be connected in either direction. To operate correctly, the sensor requires a DC voltage between 3.3V and 5V, which can be supplied through any standard interfacing method.

- Pin P1: This pin is generally connected to the +ve terminal of the power source.
- Pin P2: This pin is generally connected to GND pin of the power source.

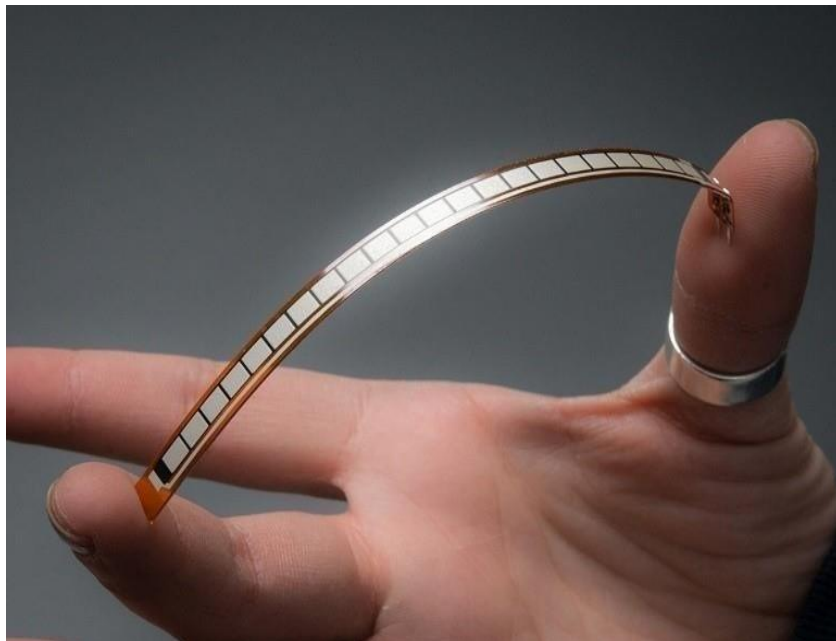


Figure 4.8: Flex Sensor

Features	Specification
Type	Resistive (Passive sensor)
Resistance (Flat)	10K ohm
Resistance (Fully Bent)	30-50K ohm
Length	2.2 inches
Thickness	0.43 mm
Operating voltage	3.3-5 V
Width	0.25 inches
Response Time	40 ms
Angle Range	0-90 degree

Table 2: Features and specification of Flex sensor

4.5.3 Accelerometer

The ADXL345 is a low-power, three-axis MEMS accelerometer that supports both I²C and SPI communication interfaces. Its Adafruit breakout boards include built-in 3.3V voltage regulation and level shifting, allowing easy connection to 5V microcontrollers such as the Arduino. Figure 4.9 shows the accelerometer module.



Figure 4.9: Accelerometer

Features	Specification
Measurement Axes	X, Y, Z (3 Axis)
Output type	Analog Voltage
Voltage supply	3.3-5 V
Output range	0.3-2.7 V per axis
Acceleration range	3g
Current consumption	350 MA
Interface to Arduino	Analog pins (A4, A5, A6, etc)

Table 3: Features and Specification of accelerometer

4.5.4 IoT Module

A Bluetooth module is a small device that enables wireless communication over short distances, operating on the 2.4 GHz frequency band. It typically uses serial communication pins, such as TX and RX, to connect with microcontrollers. Common modules include the HC-05, HC-06, and HM-10. These modules make it easy for devices like Arduino boards to communicate with smartphones or other Bluetooth-enabled devices. They offer simple pairing, stable connections, and low-power operation, making them ideal for IoT projects, robotics, and wearable devices. With Bluetooth, data can be transferred in real time without requiring an internet connection.

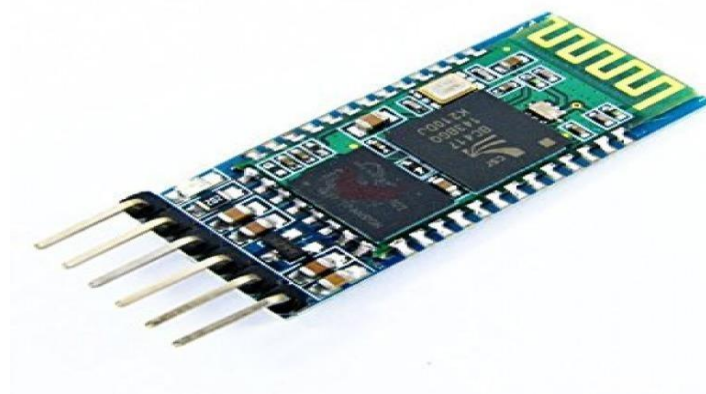


Figure 4.10: IoT Module

4.5.5 Power Supply

A power supply module is a small electronic board that converts a higher input voltage into stable, regulated outputs such as 5V or 3.3V, which are commonly needed by microcontrollers and sensors. It usually includes a DC barrel jack for an external adapter, a voltage regulator with a heat sink to maintain constant output, and capacitors to smooth any voltage fluctuations. These modules are widely used in embedded systems, Arduino projects, IoT devices, and breadboard prototypes, providing a reliable and safe power source for sensitive electronics.



Figure 4.11: Power supply

4.6 Software Requirements

Development Environment

- **Arduino IDE**
Used for programming and uploading code to the Arduino Nano.
- **Required Arduino Libraries**
Wire.h - Enables I²C communication with the MPU6050 sensor.
SoftwareSerial.h - Optional library for Bluetooth communication.
MPU6050.h or Adafruit_MPU6050.h - Libraries for interfacing with the accelerometer and gyroscope.

Smartphone Application

- **Text-to-Speech(TTS)Application**
Converts the characters received from the Arduino into audible speech for easy communication.

Tools

- **USB Cable** - For uploading code to the Arduino.
- **Serial Monitor** - Used for debugging and calibrating the sensors.

4.6.1 Arduino IDE

The Arduino Integrated Development Environment (IDE) is a software platform used to write, compile, and upload programs to Arduino boards. It includes a text editor for writing code, a message area, a console, a toolbar for common functions, and menus for various operations. Figure 4.12 shows the Arduino IDE interface.

Programs written for Arduino, called *sketches*, are saved as .ino files. The IDE supports C and C++ programming languages and provides a library from the Wiring project, which simplifies common input/output tasks. Sketches require just two basic functions: one for initializing the program and another for the main loop, which the IDE compiles and links into an executable using the included GNU toolchain.

The Arduino IDE is cross-platform and written in Java, with features like syntax highlighting, automatic indentation, brace matching, and search/replace. It allows easy one-click compiling and uploading of programs to the Arduino board. The IDE originated from the Processing and Wiring development environments and remains a simple yet powerful tool for Arduino programming.

A minimal Arduino C/C++ sketch, as seen by the Arduino IDE programmer, consist of only two functions:

- `setup()`: This function is called once when a sketch starts after power-up or reset. It is used to initialize variables, input and output pin modes, and other libraries needed in the sketch.
- `loop()`: After `setup()` has been called, function `loop()` is executed repeatedly in the main program. It controls the board until the board is powered off or is reset.

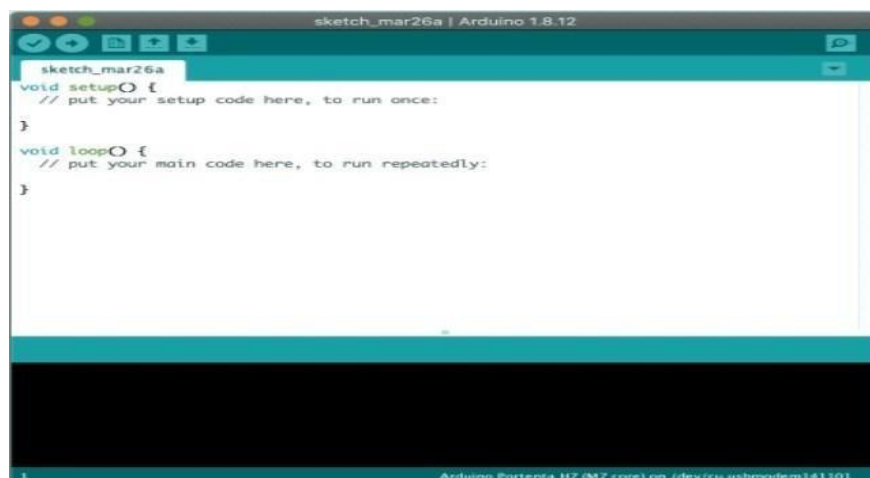


Figure 4.12: Arduino IDE

4.6.2 Mobile App

The mobile app serves as the user interface for the system. It receives processed data from the Arduino Nano, typically via Bluetooth or Wi-Fi, and can display real-time sensor readings. The app can also recognize gestures and provide feedback or alerts, making it useful for monitoring and controlling the system remotely through a smartphone. Figure 4.13 shows the mobile app interface.



Figure 4.13: Mobile App

4.6.3 Embedded C

Embedded C programming is essential for controlling microcontrollers and enabling electronic devices such as mobile phones, security systems, refrigerators, digital cameras, and more to perform specific tasks. Embedded C builds on the standard C language, using key elements like variables, data types, keywords, expressions, and functions to structure programs efficiently.

Traditionally, software for small embedded systems like the 8051 microcontroller was often written in assembly language. While assembly allows fine control over hardware and works well for small programs, it can be difficult to read, maintain, and reuse. In contrast, using a high-level language like C addresses these challenges by making code easier to understand, maintain, and reuse across projects.

Programming in C also abstracts away much of the hardware complexity. Developers do not need to fully understand the processor architecture to write functional code, and projects can be implemented more quickly. Moreover, C code is more portable between different processors, thanks to widely available ANSI C compilers.

Embedded C strikes a balance by allowing low-level hardware manipulation when needed, while providing the readability, structure, and abstraction of a higher-level language. This combination reduces development and debugging time while maintaining precise control over the hardware.

RESULTS AND ANALYSIS

5.1 Result Analysis

The Smart Glove for Sign Language Translation using IoT was successfully designed, developed, and tested to convert hand gestures into meaningful text or audio output. The prototype used flex sensors to detect finger movements and an accelerometer to track hand orientation and motion. These sensor readings were processed by the Arduino Nano to recognize predefined sign language gestures. Testing showed that the glove could accurately identify most gestures with high consistency. Each recognized gesture was transmitted wirelessly via Bluetooth to a smartphone, where it was displayed in real time on a mobile application. This demonstrated the effectiveness of IoT integration in enabling smooth and reliable communication between the wearable device and the user interface.

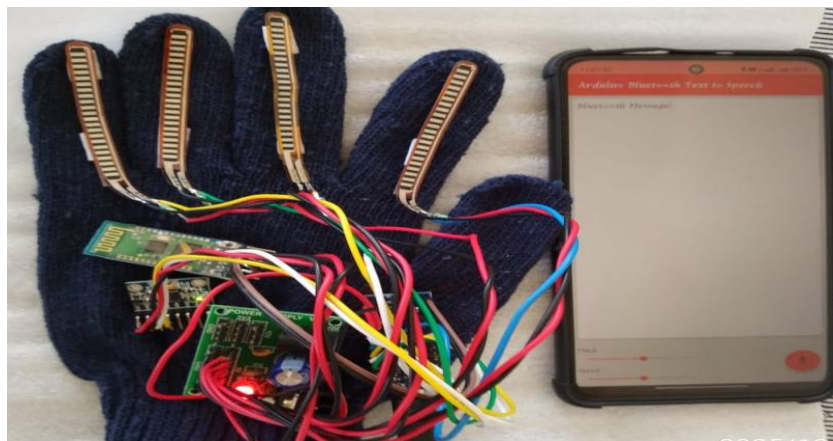
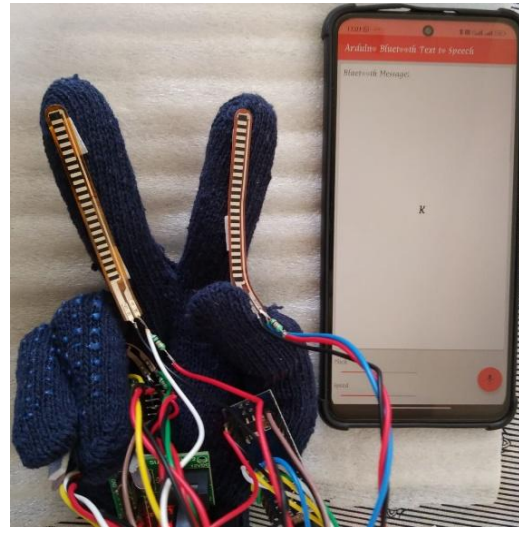


Figure 5.1: Project Setup

Figure 5.1 shows the hardware setup of a smart glove for sign language translation. The glove is fitted with multiple flex sensors stitched along the fingers, which bend according to finger movements and convert hand gestures into varying electrical signals. These signals are routed through connecting wires to an Arduino Nano, which acts as the main controller mounted on the glove. An accelerometer module is also connected to detect hand orientation and motion, helping to distinguish dynamic gestures. A Bluetooth module is integrated to enable wireless communication between the glove and a smartphone. The smartphone displayed in the image receives the processed data via Bluetooth and converts the recognized gestures into readable text or audio output.



(a)



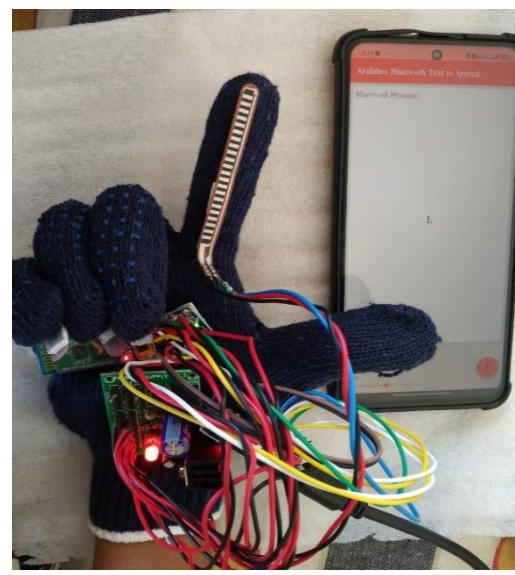
(b)

Figure 5.2: Hand Gesture “K” detection and output display

Figure 5.2 represents the ‘K’ gesture, the smart glove detects partial flexion of the middle, index, and thumb fingers using flex sensors. As the fingers bend, the sensor resistance increases from its nominal flat value of around 10 k Ω to approximately 18.2k Ω . This resistance change is converted to an analog voltage of about 2.1- 3.0 V using a voltage divider, which is then read by the Arduino Nano and mapped to finger bend angles of 35°- 50°. Additionally, the MPU6050 accelerometer detects a forward tilt along the X-axis, confirming the gesture orientation. When both the finger bend and tilt conditions are met, the system classifies the gesture as ‘K’.



(a)



(b)

Figure 5.3: Hand Gesture “L” detection and output display

Figure 5.3 represents the ‘L’ gesture, the thumb and index finger form an L-shape while the other fingers remain extended. The flex sensors on the fingers detect minimal bending, with resistances around $10.1\text{k}\Omega$ and corresponding voltage readings of 1.82–2.18 V. The finger angles range from 0° to 25° . The accelerometer detects a backward tilt along the X-axis, helping the system recognize the ‘L’ gesture reliably.

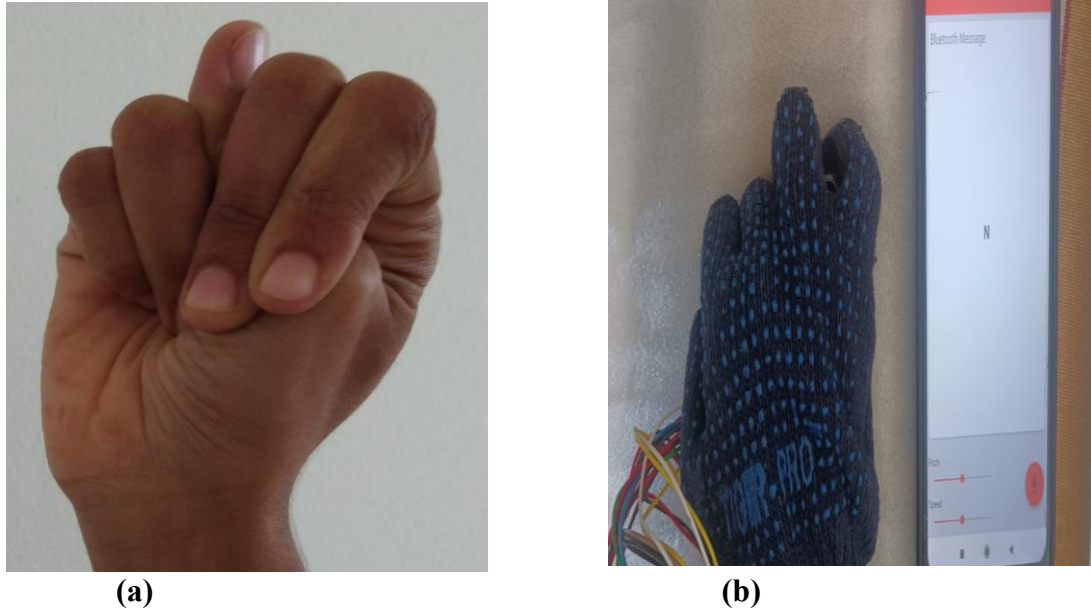


Figure 5.4: Hand Gesture “N” detection and output display

Figure 5.4 represents the ‘N’ gesture, the glove senses partial flexion of the index and middle fingers, causing the flex sensor resistance to rise to approximately $17.3\text{k}\Omega$. This produces a voltage output of 2.05-2.75 V, corresponding to finger bend angles of 30° - 45° . When both the sensor and tilt thresholds are satisfied, the gesture is classified as ‘N’.



Figure 5.5: Hand Gesture “F” detection and output display

Figure 5.5 represents the ‘F’ sign language gesture. The bending of the index finger and thumb causes the corresponding flex sensors to increase their resistance to approximately 18.4 k Ω . This resistance change is converted into an analog voltage ranging from 2.2 to 3.0 V, which corresponds to finger bend angles of about 35°- 50°. When the flex sensor thresholds and orientation conditions are satisfied, the system correctly identifies the gesture as ‘F’.

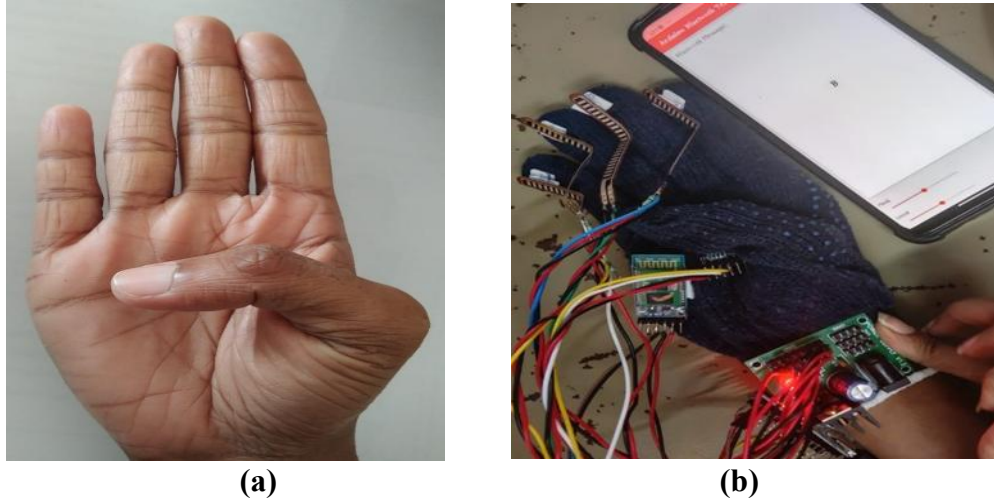


Figure 5.6: Hand Gesture “B” detection and output display

Figure 5.6 illustrates the ‘B’ sign language gesture. For this gesture, all fingers are held straight with minimal bending, resulting in flex sensor resistance values close to their nominal flat value of approximately 17.5 k Ω . This produces an output voltage in the range of 2.0 V to 2.8 V, corresponding to finger bend angles between 30° to 45°. The MPU6050 accelerometer detects a stable hand position with negligible tilt, confirming proper orientation. When these sensor readings fall within the predefined thresholds, the system classifies the gesture as ‘B’.

Gesture	Flex resistance	Voltage	Angle	Message
K	18.2	2.25-3.0	35-50	Hello! Have a nice day
L	10.1	1.82-2.18	0-25	Bye take care
N	17.3	2.05-2.75	30-45	I need water
F	18.4	2.2-3.0	35-50	Can u repeat again
B	17.5	2.0-2.8	30-45	Need to Go for Hospital

Table 4: Result table

Gesture	Expected value	Obtained value
K	18-25	18.2
L	10-15	10.1
N	17-23	17.3
F	18-25	18.4
B	17-23	17.5

Table 5: Comparison table

5.2 Summary

The developed smart glove system successfully detects hand gestures in real time. When a user performs specific gestures, the flex sensors and accelerometer accurately capture finger movements and hand orientation. The Arduino Nano processes this data and translates the gestures into the corresponding alphabet outputs. For example, the system correctly displayed the letters “K”, “L”, “N”, “F” and “B”. These results confirm that the glove can reliably recognize predefined sign language gestures and convert them into readable outputs, validating the effectiveness and accuracy of the proposed design.

CONCLUSION AND SCOPE FOR FUTURE WORK

6.1 Conclusion

The proposed system effectively combines flex sensors, an accelerometer, and IoT technology with an Arduino Nano to recognize and respond to hand gestures. Flex sensors capture finger movements, while the accelerometer tracks hand orientation, allowing the system to accurately interpret gestures and convert them into meaningful outputs, such as audio feedback through a speaker or remote updates via an IoT module. This design offers a low-cost, efficient, and portable solution suitable for applications such as assistive technology for the speech-impaired, gesture-controlled devices, and smart wearable systems. The successful implementation of this system demonstrates its strong potential for real-world applications requiring gesture-based human-machine interaction.

6.2 Scope for future work

The Smart Glove for Sign Language Translation using IoT has tremendous potential for further development. Future improvements could include integrating machine learning or deep learning to recognize a wider range of gestures, eventually supporting the full sign language vocabulary rather than just a few predefined signs. Upgrading the sensors with advanced flex sensors, pressure sensors, or bend-resistant materials could improve accuracy and durability, while a fully wireless, battery-powered design with compact embedded components would make the glove more comfortable for everyday use. Adding a built-in text-to-speech feature would allow the glove to speak directly, without relying on a mobile app, and multi-language support could make it accessible worldwide. An enhanced IoT dashboard could track user history, send alerts, and enable caregivers to monitor in real time. Future designs could also use Bluetooth Low Energy (BLE) or 5G networks for faster communication, and smart textiles or flexible circuits could make the glove washable and more flexible. By combining AI-driven gesture recognition with ergonomic wearable design, this prototype could evolve into a commercial, scalable device that greatly enhances communication for people with hearing or speech impairments.

REFERENCES

- [1] P. Mohan, M. Mohan Raj, M. Kathirvel P. A. Kasthurirngan, S. Musharaff, T. Nirmal Kumar.
“Smart Speaking Glove for Deaf and Dumb”.
- [2] Aishwarya Sharma, Dr. Siba Panda, Prof. Saurav Verma. “Sign Language to Speech Translation”.
- [3] D Abdulla, S Manav, & Jarndal A H. “Design and implementation of a sign-to speech/text system for deaf and dumb people”.
- [4] Ms. Pradnya Repal, Nikita Pore, Akanksha Pasale, Rutuja Thengal, Swati Ambule, and Prof. N.D. Deval. “Real Time Sign Language Translator Using Machine Learning”.
- [5] Abhinandan Das and colleagues. “Smart Glove for Sign Language Communications”.
- [6] Giovanni Pezzuto, Carmine Zaccagnino, and Giuseppe Valerio Lettieri. “Machine LearningBased Gesture Recognition Glove: Design and Implementation”.
- [7] Jyothi R. and Lakshmishri B. L. “An IoT-Based Smart Glove Gesture Vocalizer for Deaf and Mute People”.
- [8] Mr. Punit Vora, Mr. Bhushan Parab, Mr. Rohit Bandgar, Mr. Rahul Naik, Dr. Jyoti Dange. “MultiPurpose Smart Glove for Differently Abled Community People”.
- [9] Ahmed J Abougarair, Walaa A Arebi. “Development of a Smart Glove for Sign Language Translation Using Arduino”.
- [10] Yasir Niaz Khan, Syed Atif Mehdi. “Smart Glove for Sign Language Recognition Using Flex and Accelerometer Sensors”.
- [11] Vijay A, Lochan Pingalay S, Mohana krishnan N, Rohan Raghavendra G, Shalini Agnes Mary A. “EchoGlove: Assistive Smart Glove for Gesture-Based Communication and Health Monitoring”.
- [12] Ahmed J Abougarair, Walaa A Arebi. “Sign Language Translation Using Smart Glove and Mobile Application”
- [13] Urav Dalal Aasmi Thadhani, Mahek Upadhye, Shreya Shah, Dr. Meera Narvekar, Dr. Nilesh Patil. “Wearable Glove for Sign Language Recognition Using Flex and Gyroscope Sensors ”.
- [14] Abhishek Wahane, Rishikesh Gadade, Aditya Hundekari, Aditya Khochare, Chudaman Sukte. “Real-Time Sign Language Translation Using Wearable Sensors and Deep Learning Technique”

- [15] Urav Dalal, Aasmi Thadhani, Mahek Upadhye, Shreya Shah, Meera Narvekar, and Nilesh Patil. "Deep Learning-Enabled Smart Glove for Real-Time Sign Language Translation"
- [16] Ali, S. Munawwar, and B. Nadeem. "Electronic Speaking Glove for Speechless Patients".
- [17] M. Wald. "Captioning for Deaf and Hard of Hearing People by Editing Automatic Speech Recognition in Real Time".
- [18] N. P. Bhatti, A. Baqai, B. S. Chowdhry, and M. A. Unar. "Electronic Hand Glove for Speech Impaired and Paralyzed Patients".
- [19] Christopher Harris and Samantha Carter. "Design and Development of a Smart Glove for Assistive Communication".
- [20] Daniel Robinson and Rachel White. "Integration of Machine Learning Algorithms in Gesture Recognition for Assistive Devices".

APPENDIX A

Arduino IDE: The Arduino IDE is an open-source software used for writing and uploading code to Arduino boards. It works on multiple operating systems, including Windows, Mac OS X, and Linux, and supports programming in C and C++. IDE stands for Integrated Development Environment, providing a complete workspace for coding, compiling, and uploading programs. Writing a program in the Arduino IDE is called *sketching*, and sketches are saved with the .ino file extension. To upload a sketch, the Arduino board must be connected to the IDE, allowing the code to be transferred and run on the hardware.

Arduino Library: In Arduino, a library is an advanced feature that extends the capabilities of the IDE by providing additional functionality. Libraries are typically written in C or C++ (.cpp) and allow programmers to easily work with hardware and manipulate data without writing complex code from scratch. To use a library in the Arduino IDE, you can go to Sketch Import Library. Many libraries are available for download, and it is also possible to create custom libraries to suit specific project needs.

APPENDIX B

```
#include <Adafruit_MPU6050.h>
#include <Adafruit_Sensor.h>
#include <Wire.h>
Adafruit_MPU6050 mpu;
void setup(void)
{
  Serial.begin(9600);
  while (! Serial)
    delay(10); // will pause Zero, Leonardo, etc until serial console opens
  Serial.println("Hand Gesture Recognition");
  // Try to initialize!
  if (!mpu.begin())
  {
    Serial.println("Failed to find MPU6050 chip");
    while (1)
    {
      delay(10);
    }
  }
  // Serial.println("MPU6050 Found!");
  mpu.setAccelerometerRange(MPU6050_RANGE_8_G);
  // Serial.print("Accelerometer range set to: ");
  switch (mpu.getAccelerometerRange())
  {
    case MPU6050_RANGE_2_G:
      // Serial.println("+2G");
      break;
    case MPU6050_RANGE_4_G:
      // Serial.println("+4G");
      break;
    case MPU6050_RANGE_8_G:
      // Serial.println("+8G");
```

```

break;

case MPU6050_RANGE_16_G:
//  Serial.println("+16G");
break;
}

mpu.setGyroRange(MPU6050_RANGE_500_DEG);
//  Serial.print("Gyro range set to: ");
switch (mpu.getGyroRange())
{
case MPU6050_RANGE_250_DEG:
//  Serial.println("+ 250 deg/s");
break;
case MPU6050_RANGE_500_DEG:
//  Serial.println("+ 500 deg/s");
break;
case MPU6050_RANGE_1000_DEG:
//  Serial.println("+ 1000 deg/s");
break;
CaseMPU6050_RANGE_2000_DEG:
//  Serial.println("+ 2000 deg/s");
break;
}

mpu.setFilterBandwidth(MPU6050_BAND_5_HZ);
//  Serial.print("Filter bandwidth set to: ");
switch (mpu.getFilterBandwidth())
{
case MPU6050_BAND_260_HZ:
//  Serial.println("260 Hz");
break;
case MPU6050_BAND_184_HZ:
//  Serial.println("184 Hz");
break;
case MPU6050_BAND_94_HZ:
//  Serial.println("94 Hz");

```

```

break;
case MPU6050_BAND_44_HZ:
//  Serial.println("44 Hz"); break;
case MPU6050_BAND_21_HZ:
//  Serial.println("21 Hz"); break;
case MPU6050_BAND_10_HZ:
//  Serial.println("10 Hz"); break;
case MPU6050_BAND_5_HZ:
//  Serial.println("5 Hz");
break;
}
//  Serial.println(""); delay(100);
}
void loop()
{
//  sensors_event_t a, g, temp;
//  mpu.getEvent(&a, &g, &temp);
//  /* Print out the values */
//  Serial.print("Acceleration X: ");
//  Serial.print(a.acceleration.x);
//  Serial.print(", Y: ");
//  Serial.print(a.acceleration.y);
//  Serial.print(", Z: ");
//  Serial.print(a.acceleration.z);
//  Serial.println(" m/s^2");
//  Serial.print("Rotation X: ");
//  Serial.print(g.gyro.x);
//  Serial.print(", Y: ");
//  Serial.print(g.gyro.y);
//  Serial.print(", Z: ");
//  Serial.print(g.gyro.z);
//  Serial.println(" rad/s");
//  Serial.print("Temperature: ");
//  Serial.print(temp.temperature);

```

```

// Serial.println(" degC");
// Serial.println("");
// delay(1000);
// int flux1val=analogRead(A0);
// int flux2val=analogRead(A1);
// int flux3val=analogRead(A2);
// int flux4val=analogRead(A3);
// Serial.println("Flux_1:"+String(flux1val));
// Serial.println("Flux_2:"+String(flux2val));
//Serial.println("Flux_3:"+String(flux3val));
//Serial.println("Flux_4:"+String(flux4val));
// delay(1000);
sensors_event_t a, g, temp; mpu.getEvent(&a, &g, &temp);
float ax = a.acceleration.x;
float ay = a.acceleration.y;
float az = a.acceleration.z;
// Serial.print("Acceleration X: ");
// Serial.print(ax);
// Serial.print(" Y: ");
// Serial.print(ay);
// Serial.print(" Z: ");
// Serial.println(az);
float threshold = 3.0; // Sensitivity for tilt detection
if (ax > threshold && abs(ay) < threshold)
{
Serial.println('K');
delay (1200);
}
else if (ax < -threshold && abs(ay) < threshold)
{
Serial.println('L');
delay (1200);
}

```

```

else if (ay > threshold && abs(ax) < threshold)
{
Serial.println('M');
delay(1200);
}
else if (ay < -threshold && abs(ax) < threshold)
{
Serial.println('N');
delay(1200);
}
Else
{
// Serial.println("Stable");
// Optional: when no clear tilt
}
// Serial.println();
// ---- Flux sensors (unchanged) ----
int flux1val = analogRead(A0);
int flux2val = analogRead(A1);
int flux3val = analogRead(A2);
int flux4val = analogRead(A3);
// Serial.println("Flux_1:" + String(flux1val));
// Serial.println("Flux_2:" + String(flux2val));
// Serial.println("Flux_3:" + String(flux3val));
// Serial.println("Flux_4:" + String(flux4val));
// Serial.println("-----");
delay(1000);
if(flux1val<400)
{
// Serial.println("Can u repeat again");
Serial.println('A');
delay(1200);
}
if(flux2val<400)

```

```

{
//  Serial.println("Need to Go for Hospital");
Serial.println('B');
  delay(1200);
}
if(flux3val<400)
{
//  Serial.println("Hello!! Have a nice day");
Serial.println('C');
delay(1200);
}
if(flux4val<400)
{
//  Serial.println("Bye Takecare");
Serial.println('D');
delay(1200);
}
if(flux1val<400 && flux2val<400)
{
Serial.println('E'); delay(1200);
}
if(flux1val<400 && flux3val<400)
{
Serial.println('F'); delay(1200);
}
if(flux1val<400 && flux4val<400)
{
Serial.println('G'); delay(1200);
}
if(flux2val<400 && flux3val<400)
{
Serial.println('H');
delay(1200);
}
}

```

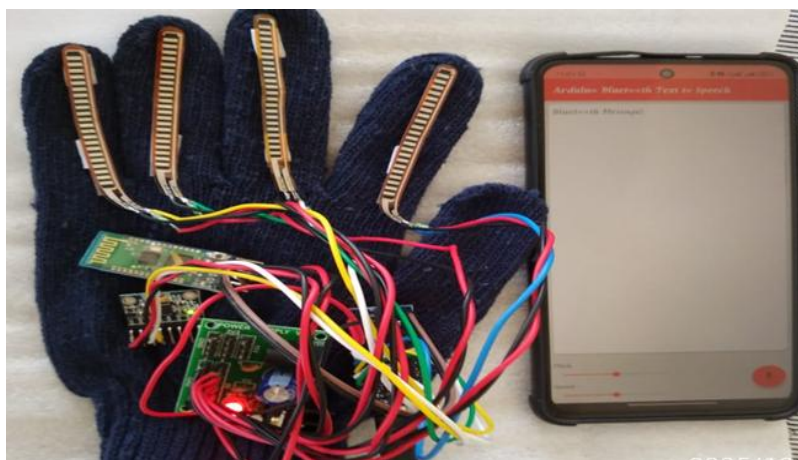
```

if(flux2val<400 && flux4val<400)
{
Serial.println('I');
delay(1200);
}
if(flux3val<400 &&flux4val<400)
{
Serial.println('J');
delay(1200);
}
}
}

```



Project Demonstration by the Team



Project Model



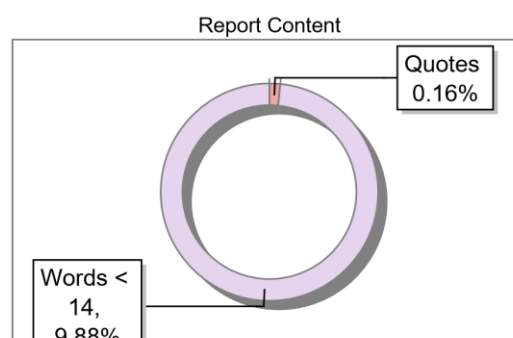
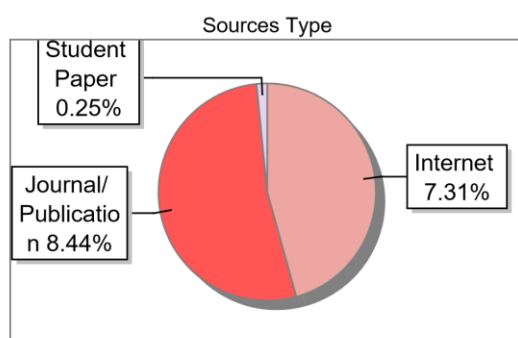
The Report is Generated by DrillBit Plagiarism Detection Software

Submission Information

Author Name	rachana
Title	smart
Paper/Submission ID	5015005
Submitted by	ece@sdmit.in
Submission Date	2025-12-19 17:08:17
Total Pages, Total Words	21, 3653
Document type	Project Work

Result Information

Similarity **16 %**



Exclude Information

Quotes	Not Excluded	Language	English
References/Bibliography	Not Excluded	Student Papers	Yes
Source: Excluded < 14 Words	Not Excluded	Journals & publishers	Yes
Excluded Source	0 %	Internet or Web	Yes
Excluded Phrases	Not Excluded	Institution Repository	Yes

Database Selection

A Unique QR Code use to View/Download/Share Pdf File



PERSONAL PROFILE

	Mrs. Shaileshwari S is an Assistant Professor in the Department of Electronics and Communication Engineering at SDM Institute of Technology, Ujire, with more than 12 years of teaching experience. She obtained her M. Tech degree in VLSI Design and Embedded Systems from KVGCE College of Engineering, Sullia, and B.E. in Electronics and Communication Engineering from SDM Institute of Technology, Ujire. Her areas of interest include Digital System Design, Analog Circuits, Communication Engineering, and Linear Integrated Circuits.
	Name: Rachana Jalisatgi USN: 4SU22EC074 Address: D/o: Rajesh V Jalisatgi, Udyam nagar, Honnavar Taluk, Uttara Kannada district-581334. E-mail ID: rachanajalisatgi@gmail.com Contact Phone No: 8088570595
	Name: Rashmi Shetty USN: 4SU22EC082 Address: D/o: Ramesh Shetty, Guripalla post, Belthangady Taluk, Dakshina Kannada district- 574240 E-mail ID: rakshmishetty562@gmail.com Contact Phone No: 6364728745

